



CIS-3321-40 Network Administration

Meeting Days, Times, Location: T/TH 6:00 PM - 7:15 PM

Course Modality: Hybrid

Semester: Fall, **Year:** 2024

Section(s): CIS-3321-40

Instructor Information

Name: William Nash

Contact Information: (972)372-4557

Office hours: By Appointment

Course Introduction

Welcome to Introduction to Computer Networking! This comprehensive course will introduce you to the fundamentals of computer networking, preparing you for the CompTIA Network+ certification exam. We'll explore network technologies, installation and configuration, media and topologies, management, and security. Through a combination of lectures, video content, hands-on labs, and practical assignments, you'll gain the knowledge and skills necessary to design, configure, manage, and troubleshoot both wired and wireless networks. Whether you're looking to start a career in IT or enhance your existing skills, this course will provide you with a solid foundation in networking principles and practices.

Course Description

This course provides a comprehensive introduction to computer networking concepts and practices. It covers network technologies, installation and configuration, media and topologies, management, and security. The course prepares students for the CompTIA Network+ certification exam and includes hands-on lab exercises to reinforce theoretical concepts.

Course Learning Objectives	Degree Program or GEC Competency Goals
Understanding and describing the fundamental concepts of computer networking	CIS
Identify and explain various network technologies and topologies	CIS
Configure and troubleshoot network devices and protocols	CIS
Implement network security measures and best practices	CIS
Prepare for and successfully complete the CompTIA	CIS

Course Learning Objectives	Degree Program or GEC Competency Goals
Network+ certification exam	

Required Materials

Textbook(s)

Author: Todd Lammle
 Title: CompTIA Network+ Study Guide: Exam N10-008
 Edition: 5th Edition
 Publisher: Sybex
 Year: 2021
 ISBN: 978-1119811626

Software/Apps/Equipment

- Packet Tracer or similar network simulation tool
- Wireshark
- Access to a computer with admin rights for installing required software

Instruction/Learning Methods and Technologies

The learning methods used in this course include the following:

- x Discussion forums and/or wikis
- x Synchronous (real-time) virtual meetings with peers and/or the instructor
- x Streaming video
- x Quizzes
- x Exams
- x Collaborative assignments
- x Independent homework assignments
- x Writing-intensive assignments
- x Clinical or fieldwork
- x Other: Hands-on lab exercises using network simulation tools

Student Workload Expectation

Students should expect to spend approximately 9-12 hours per week on this course, including lecture time, reading assignments, lab exercises, and independent study. This workload may vary depending on individual learning styles and backgrounds.

Course Requirements

Submitting and Returning Assignments

Assignment Submission

- Submit all of your assignments via Canvas.
- Use the standard Microsoft® Office programs such as Word, PowerPoint, and Excel (available to you through your txwes.edu Office 365 account) to complete your assignments unless assignment directions specify that you should use a different file format.
- Use the following guidelines when formatting your documents:
 - 12-point font, double-spaced
 - Include your name, date, and assignment title on all submissions

Assignment Feedback

- I will provide feedback on assessments within seven days after submission.
- My feedback will be provided as comments in Canvas and/or rubric evaluations.
- Your grades will be posted to the Canvas Gradebook.

Communication with the Instructor

- If you have questions or concerns about this course, please contact me. I will provide feedback on assessments within seven days of submission.
 - My feedback will be provided as comments in Canvas and rubric evaluations.
 - Your grades will be posted to the Canvas Gradebook.

Please email me with questions of a personal nature using either your University email account or the email system in Canvas. I will respond to emails within 24 hours.

Quizzes and Exams

Quizzes and exams will be administered through Canvas or at the Computerminds testing center. 112 South Ector Dr. Euless, TX 76040 Ph: (817) 858-9670
Some exams may require the use of the Respondus Lockdown Browser. Please ensure you have this installed on your computer before the exam date.

Evaluation & Grading

All grades will be made available to students in Canvas.

Assignment Category	Number of Graded Items	Point Value per Item	Total Points
Discussions	14	20	280
Assignments	14	50	700
Quizzes	14	25	350
Exams	2	670	670
Total Course Points:			2000

Final Grade	Final Points/Average
A	1800-2000
B	1600-1799
C	1400-1599
D	1200-1399
F	0-1199

Academic Integrity

- Cheating, plagiarism (submitting another person's material as one's own), or completing assignments for another person who will receive academic credit are not permitted. This includes the use of unauthorized books, notebooks, or other sources in order to secure or give help during an examination, the unauthorized copying of examinations, assignments, reports, or term papers, or the presentation of unacknowledged material as if it were the student's own work. Disciplinary action may be taken beyond the academic discipline administered by the course instructor. [Read the University's Academic Integrity Policy.](#)

Class Attendance Policy

Hybrid

Students are expected to attend all scheduled classes regularly and punctually. Students are **required** to notify instructors prior to any missed class and are responsible for all class and laboratory assignments.

Online and Hybrid Class

Students enrolled in distance education courses must demonstrate **academic engagement** throughout the course to be counted as present.

- Academic engagement means actively engaging with course content.** Examples of academic engagement include attending a synchronous session, interacting with the instructor, participating in an online discussion, completing interactive tutorials or web-based assignments, and completing assignments or exams.

- Academic engagement **does not include** the act of logging into Canvas, nor does it include participating in academic counseling or advising.

Students must demonstrate academic engagement a minimum of once each week in order to be marked present for that week. During the first week of class, all students will be required to complete either a grade-based or submission-based verification of academic engagement. Students are expected to attend all scheduled face-to-face sessions and actively participate in online activities. If you must miss a face-to-face session, please notify the instructor in advance and make arrangements to obtain the missed information from a classmate.

University Authorized Absences

- University regulations authorize certain absences of students when representing the University (e.g. participation in intercollegiate athletic competitions, and student government, student development, or fine arts events).
- Absences that may be necessary to fulfill course requirements and are approved by the Provost are also considered to be authorized.
- Students are required to notify instructors prior to any missed class and will be held responsible for all class and laboratory assignments.

Faculty-Initiated Student Withdrawal from a Course

When a student exceeds the permissible number of absences for a class, or, fails to meet the course participation expectations for a class the instructor may drop the student from the class using the Faculty Initiated Student Withdrawal process.

The Registrar will record a grade of "DP" for the course on the student's transcript. The Registrar will record a grade of "WF" (failure) on the student's transcript if the instructor drops the student after the date for receiving a grade of "DP" as specified in the University catalog.

*Note: Per the Office of Financial Aid, students who withdraw from a course **cannot** be re-enrolled in that course during the same semester.*

[The complete Attendance Policy is available in the Catalog.](#)

Other University Policies

- Selected University policies are available in the [Canvas TXWES STUDENT HUB](#). The current Texas Wesleyan University [Catalog](#) and [Student Handbook](#) contain all University policies.

Note: Course syllabi are intended to provide students with basic information concerning the course. The syllabus can be viewed as a "blueprint" for the course. Instructors will inform students of any changes concerning examinations, the grading or attendance policies, or changes in project assignments.

[Syllabi for courses involving laboratories which use chemicals should include the following:](#)

Due to the limitless variety of potentially hazardous chemicals that may be found in a scientific or other laboratory environment, students having known, or potential, health-related concerns, such as allergies, asthma, contact dermatitis, pregnancy, or other physiological sensitivities should check with their physician on the advisability of laboratory work. The instructor and/or chair of the teaching department will provide, upon request, a list of chemicals that will be used in experiments during the semester. More detailed information (that is, Materials Safety Data Sheets) will be furnished by the deans upon written

request.

Schedule

Week	Topic	Learning Activities	Assessments/Due Dates
1	Introduction to Networking	Read: Chapter 1, Watch: Professor Messer videos	Discussion Board (Tues), Quiz (Thursday)
2	Network Infrastructure and Hardware	Read: Chapter 2, Watch: Professor Messer videos	Lab Assignment (Tuesday), Quiz (Thursday)
3	Network Protocols and Services	Read: Chapters 3-4, Watch: Professor Messer videos	Discussion Board (Tues), Lab Assignment (Tues), Quiz (Thursday)
4	Wireless Networking	Read: Chapter 5, Watch: Professor Messer videos	Lab Assignment (Tuesday), Quiz (Thursday)
5	Network Security Fundamentals	Read: Chapters 6-7, Watch: Professor Messer videos	Discussion Board (Tues), Lab Assignment (Tues), Quiz (Thursday)
6	Network Troubleshooting	Read: Chapter 8, Watch: Professor Messer videos	Lab Assignment (Tuesday), Quiz (Thursday)
7	Cloud Computing and Virtualization	Read: Chapter 9, Watch: Professor Messer videos	Discussion Board (Tues), Lab Assignment (Tues), Quiz (Thursday)
8	Midterm Review and Assessment	Review all previous materials	Midterm Exam
9	Advanced Network Security	Read: Additional materials, Watch: Professor Messer videos	Discussion Board (Tues), Lab Assignment (Tues), Quiz (Thursday)
10	Network Performance and Optimization	Read: Chapter 10, Watch: Professor Messer videos	Lab Assignment (Tuesday), Quiz (Thursday)
11	Business Continuity and Disaster Recovery	Read: Chapter 11, Watch: Professor Messer videos	Discussion Board (Tues), Lab Assignment (Tues), Quiz (Thursday)
12	Mobile and Remote	Read: Chapter 12,	Lab Assignment

	Networking	Watch: Professor Messer videos	(Tuesday), Quiz (Thursday)
13	Network Automation and Programmability	Read: Chapter 13, Watch: Professor Messer videos	Discussion Board (Tues), Lab Assignment (Tues), Quiz (Thursday)
14	Emerging Network Technologies	Read: Additional materials, Watch: Professor Messer videos	Lab Assignment (Tuesday), Quiz (Thursday)
15	CompTIA Network+ Exam Preparation	Review all materials	Discussion Board (Tues), CompTIA Network+ Practice Exam (Sun)
16	Final Exam	Complete Final Exam	Schedule your final exam at Computerminds.

This course provides a comprehensive introduction to computer networking, covering both theoretical concepts and practical skills. The course is structured around the following major units:

1. Fundamentals of Networking: Basic concepts, OSI and TCP/IP models, network topologies
2. Network Hardware and Infrastructure: Routers, switches, cabling, and network design
3. Network Protocols and Services: IP addressing, routing protocols, DNS, DHCP
4. Wireless Networking: Wi-Fi standards, security, and implementation
5. Network Security: Threats, encryption, firewalls, and best practices
6. Troubleshooting and Performance Optimization: Common issues, tools, and techniques
7. Cloud Computing and Virtualization: Concepts, implementation, and management
8. Emerging Technologies: SDN, IoT, 5G, and their impact on networking

Throughout the course, students will engage in hands-on lab exercises using network simulation tools to reinforce theoretical concepts. The course also prepares students for the CompTIA Network+ certification exam, covering all major exam objectives.

Detailed Assignment Descriptions

Discussion Board Posts

Each week, students will participate in online discussions related to the current topic. Posts should be 250-300 words and demonstrate critical thinking and application of course concepts. Students are also required to respond to at least two peers with substantive comments.

Lab Assignments

Lab assignments will involve hands-on practice with networking concepts using tools like Packet Tracer or GNS3. These assignments will typically include tasks such as network configuration, troubleshooting

scenarios, and security implementations. Detailed instructions and rubrics will be provided for each lab assignment.

Quizzes

Weekly quizzes will assess understanding of key concepts from the readings and lectures. Quizzes will consist of multiple-choice and short answer questions and must be completed within the specified time limit.

Midterm Exam

The midterm exam will cover material from the first half of the course. It will include a combination of multiple-choice questions, short answer questions, and a practical component requiring analysis of network scenarios.

CompTIA Network+ Practice Exam

This comprehensive practice exam will simulate the actual CompTIA Network+ certification exam, helping students assess their readiness and identify areas for further study.

Support Services

Texas Wesleyan University offers various support services to help students succeed:

1. IT Help Desk: For technical issues related to university systems, email helpdesk@txwes.edu or call (817) 531-4428.
2. Academic Success Center: Offers tutoring services for various subjects. Visit <https://txwes.edu/academics/success-center/> for more information.
3. West Library: Provides research assistance and resources. Visit <https://txwes.edu/library/> or contact a librarian at library@txwes.edu.
4. Counseling Center: Offers mental health support and counseling. For appointments, call (817) 531-4859.
5. Career Services: Provides career counseling and job search assistance. Visit <https://txwes.edu/career-services/> for more information.
6. Accessibility Services: If you require accommodations due to a disability, contact the Office of Accessibility Services at accessibility@txwes.edu.

Students are encouraged to take advantage of these services throughout the semester to support their academic success.

Diversity and Inclusion Statement

In this course, we are committed to creating an inclusive learning environment that values diversity of thought, backgrounds, and experiences. We believe that diversity enriches the learning experience for all students and prepares you for careers in an increasingly diverse tech industry. All students are expected to treat their peers and instructor with respect, regardless of race, ethnicity, gender identity, sexual orientation, socioeconomic background, age, disability, or religion. If you have any concerns about inclusion or respectful treatment in the course, please speak with the instructor or refer to the university's diversity and inclusion policies.

Career Relevance

The skills and knowledge gained in this course are directly applicable to various careers in IT and networking. Some potential career paths include:

1. Network Administrator
2. Systems Administrator
3. Network Security Specialist
4. Cloud Network Engineer
5. IT Support Specialist
6. Network Architect

Additionally, the CompTIA Network+ certification that this course prepares you for is widely recognized in the industry and can enhance your employability and career prospects.

Throughout the course, we will discuss how specific topics relate to real-world job roles and industry trends. Guest speakers from the IT industry may be invited to provide insights into career opportunities and the practical application of course concepts.